How can cybersecurity and data privacy help build digital trust in the future of embedded finance in the UAE?

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I. Introduction

Technology has changed the way we all communicate, the way we shop, the way we study and the way we go about our day to day lives. Finances are no different. We can now pay via our smartphones, move money around via apps and access our banking without ever having to go into a branch. It's all become digital and that has in turn created a whole host of new opportunities for businesses to innovate in the way they manage payments and financial services.

One of the developments as part of this wider digital transformation is embedded finance. Embedded finance is the integration of financial services into non-financial platforms, for example, a customer shopping online could potentially make a payment directly through the shopping app rather than having to open up a separate banking app. Some platforms will also offer integrated payment plans, insurance and other financial products. The end result is that the customer can access financial services within the application they're already using.

Despite the convenience of embedded finance, it does raise issues around cyber security, data privacy and digital trust. Financial services are inherently dealing with sensitive information (personal data, transactional data etc). When multiple organisations and systems are working together to deliver a service the issue of protecting that data can become considerably more complex. A vulnerability in one part of a connected system can pose a risk to the other participants. Customers may also be left in the dark as to which entities have access to their data and how it will be used.

Cybersecurity is about protecting computer systems, networks and data from digital threats and unauthorised access. Data privacy is about how personal information is collected, processed, shared and stored. Digital trust is the degree of confidence that people have in digital services and the organisations that deliver them. The three are very much interlinked as people will be less likely to use a financial service if they are concerned about the security of their money or their own personal data.

The UAE is a pertinent case in which to look at these issues. It's in the process of building out its digital financial environment and the UAE Central Bank has introduced an Open Finance framework which is designed to enable secure data sharing and financial innovation. That in itself provides a good opportunity to look at how technological safeguards and regulation can be used to underpin the responsible development of financial services.

The objective of this research is to explore how cybersecurity and data privacy can be leveraged to foster digital trust in the future of embedded finance in the UAE.

So:

How can cyber security and data privacy actually help to build digital trust in the future of embedded finance in the UAE?

The research is based on secondary sources (UAE regulatory publications, cybersecurity guidance, research into digital trust etc). It looks at how embedded finance works, identifies the potential security and privacy risks, looks at the relevant UAE regulatory regime and considers some practical examples. The aim is to understand the issues involved and to suggest some ways in which financial technology providers can go about protecting their users as they continue to develop these increasingly convenient digital services.

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1. https://centralbank.ae/en/our-operations/fintech-digital-transformation/open-finance/
   (official text consent-based sharing of financial data and the UAE's approach to digital financial services)
2. https://rulebook.centralbank.ae/en/rulebook/open-finance-regulation
   (official regulatory background for secure data sharing and financial services)

II. Understanding Embedded Finance

Embedded finance refers to the distribution of financial services in digital platforms that are not necessarily financial institutions. In most cases, people would traditionally go to a bank, payment provider, or other financial entities to perform a financial service. However, embedded finance offers certain financial services directly in an alternate platform and makes the overall process more convenient for the customer.

For instance, when a customer shops on an online shopping application, they might be offered the ability to pay for the item with a payment service at checkout. The customer does not need to navigate to a separate website to make the purchase. Some platforms even facilitate instalment plans during checkout or provide insurance as part of their services. Those are examples of embedded finance. Embedded finance should also not be confused with digital banking, as the latter refers to banks' presence on digital space, such as the website or

mobile applications provided by the banks themselves. While similar, embedded finance and digital banking are two separate concepts.

The concept of open finance is also connected to embedded finance. Open finance refers to the availability of financial information and, crucially, the ability for the customer to authorise a financial institution to launch a specific transaction. In some ways, it facilitates embedded finance by allowing different financial services to share information and for customers to use different products with different financial institutions, if they so desire. With that said, not all examples of embedded finance are necessarily a part of open finance, since the two operate as separate concepts.

A crucial element of most examples of embedded finance is the application programming interface (API). An API is a set of programming instructions that allows two software programs to communicate with each limited set of instructions. For instance, the online shopping website might be using an API to send a request to a payment provider to process a customer's payment. The payment provider then responds with either a positive or negative response.

APIs allow organizations to utilise the tools developed by other companies without building every single element in-house and help to make embedded finance possible. They generally make the process easier for businesses, as well as ordinary consumers. With that said, APIs are generally not secure, and a wrong set of instructions or permissions might make sensitive information public or allow unauthorised transactions.

Embedded finance, and particularly the use of APIs, has several advantages. It generally makes financial transactions easier and shorter, as well as make them more accessible, since they are available in popular digital platforms. Additionally, businesses can offer financial options without necessarily developing an entire system for financial services. However, the concept also has several disadvantages. First and foremost, it is often not obvious what company is responsible for a particular financial service, and those that do embed finance services might be taking advantage of the trust of the bigger company. Additionally, the involvement of multiple entities in a transaction increases the likelihood of fraud and customer complaints and makes regulatory control and consumer protection more difficult. The convenience of embedded finance does not automatically translate into security and accessibility for the consumer.

The UAE's Open Finance framework essentially refers to a specific set of rules designed for embedded finance in the United Arab Emirates. The Central Bank

of the UAE has established certain requirements for organizations that wish to participate in the system, which operate as a set of rules for all entities. As a rule, the UAE's Open Finance framework focuses on protecting the consumer by ensuring that appropriate security measures such as consent, authentication, communication encryption, and technology-related risks are addressed.

Overall, embedded finance represents a major shift towards greater convenience for the customers. However, that convenience comes at the cost of increased likelihood of fraud and consumer issues, which are why it is important for organisations to understand their role in the system and operate as responsible parties. This is why understanding the technology behind APIs and embedded finance is a necessary step before moving on to the discussion of cybersecurity and consumer protection.

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3. https://centralbank.ae/en/our-operations/fintech-digital-transformation/open-finance/
   (helps explain how financial services can connect with other digital services through consent-based data sharing)

III. Cybersecurity and Data Privacy in Embedded Finance

Cybersecurity and data privacy are closely-related but ultimately different fields which concern the protection of users and their information. The former involves keeping systems, networks, apps, and data safe from digital attacks, while the latter deals with the collection, use, sharing, and storage of personal information. Since embedded finance typically involves the processing of data through multiple digital platforms and services, both fields are of paramount importance for such operations.

Firstly, embedded finance operations are highly susceptible to breaches and cyberattacks, in particular, to unauthorised access and disclosure. In general, the issue arises due to unsuitable or unencrypted passwords, unauthorised access to login details, incorrect permission settings, and the absence of identity verification. Unauthorised access may provide attackers with access to users' personal details, while in the case of financial services, such breaches also endanger financial assets. One of the ways to prevent unauthorised access is to implement authentication mechanisms. Authentication refers to verifying that a user is authorised to access a particular domain, system, or application. For example, the application may ask to enter a password or turn on notifications to

confirm that the user is legitimate. Applications may also implement multi-factor authentication, which is typically more secure than single-factor authentication.

Secondly, the issue of API security is of paramount importance in the context of embedded finance. For example, an application should ensure that only users who have the requisite permissions can view a specific record or file. Developers should also implement validation mechanisms to ensure that requests to the application are authorised and originate from trusted sources. The same applies to the protection of data in transit and at rest: in particular, the use of encryption tools is advisable to prevent would-be-attackers from utilising information, should they manage to intercept it.

Beyond that, the threat of fraud and social engineering cannot be ignored. Cyber attackers may impersonate banks and other financial organisations, tricking users into providing bank details or authentication codes. From the customer's perspective, the issue is addressed through the implementation of appropriate customer communication and education regarding dubious requests for sensitive information. On the developer side, the issue may be addressed through the implementation of appropriate authentication and verification tools.

Connected finance platforms typically entail the use of multiple software solutions, including third-party components. As a result, organisations should invest more effort in ensuring that such components are up-to-date and free from vulnerabilities, scanning the ecosystem for potential weaknesses and threats. In addition, firms should also invest in developing and implementing appropriate incident response plans, clarifying interdependent obligations, and outlining the steps to be taken in case of an attack.

In general, the Central Bank of the UAE's Open Finance Regulation provides the essential requirements in terms of technology risk management, cybersecurity, secure communication, and authentication tools, as well as other relevant criteria. In general, the regulation says that connected finance operations should have appropriate security mechanisms in place, which goes beyond the responsibility of customers in terms of protecting their personal information.

The National Institute of Standards and Technology's Cybersecurity Framework 2.0 presents a useful organisational structure in terms of cybersecurity management. Its framework divides the entire lifecycle of managing cybersecurity risks into six steps: Govern, Identify, Protect, Detect, Respond, and Recover. In short, the framework suggests that organisations must appoint

the relevant personnel to oversee and manage cybersecurity risks and identify the key threats and vulnerabilities. They should implement tools and procedures to prevent attacks, identify and analyse incidents, and finally, develop a series of responsive procedures.

As for data privacy, despite the presence of robust cybersecurity measures, personal information and data should still be adequately protected to prevent unauthorised use. Financial institutions must invest more effort in designing privacy-positive practices and procedures which would help minimise the risks associated with personal data processing and adhere to the relevant laws and regulations.

Embedded finance platforms and applications should seek to implement informed consent practices and procedures, enabling customers to have appropriate control and visibility over their data. In this instance, the most important tenet is to ensure that users provide genuine informed consent: in other words, organisations should refrain from disguising the terms and conditions of data processing behind overly complicated legal language. At the same time, the principle of informed consent does not override the other aspects of privacy law and regulation: in particular, not all aspects of data processing require consent.

Another important principle is data minimisation, which refers to the organisation's ability to collect and use only the necessary data. For example, in the case of payment processing, the organisation utilises the customer's sensitive bank details, as without them, the payment would not be completed. However, the organisation does not have the right to utilise the information for purposes for which it is not intended.

Another important principle is data retention: in short, the organisation should only retain the relevant information for the appropriate period of time. If possible, the personal data should be anonymised or deleted upon the expiry of the retention period. In any case, the organisation should ensure that the retention of personal data is appropriate and does not present any undue risks.

Transparency is another key principle, without which customers would not be able to fully understand how their data is processed. In this instance, the organisation should provide all the relevant information in an accessible manner: in other words, it should not be instructed via overly complicated language which would require a specialist to interpret. In the context of sharing information with third parties, the customers should also be provided with appropriate information as to how such organisations process their data. In

short, the data privacy-related aspects should be clearly outlined in a privacy notice.

The UAE has a robust legal and regulatory framework concerning the protection of data privacy, which is primarily governed by the Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data. In particular, the law stipulates the basic principles, which data controllers should follow when processing information. However, the implementation of such principles is subject to other laws and regulations, including those pertaining to the free zones, the relevant sector-specific laws, and the regulations issued by the Central Bank of the UAE.

In general, the key challenge that the financial institutions face in terms of privacy and security is the complexity of managing the security and privacy aspects on the one hand, and the ability to provide a seamless experience for customers on the other. There is a tension between the requirement to obtain informed consents from customers and the ability to avoid disrupting the purchasing or the financial experience unnecessarily.

It is worth noting that both privacy and cybersecurity threats will never be eliminated: new threats and challenges will constantly appear. However, the implementation of appropriate consent, authentication, and other procedures will help to mitigate some of the risks. In particular, the use of appropriate authentication mechanisms will help to prevent unauthorised access to users' data and financial resources, which is of paramount importance if the services in question store or process sensitive information.

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4. https://rulebook.centralbank.ae/en/rulebook/open-finance-regulation
5. https://rulebook.centralbank.ae/en/entiresection/6170
   (cyber-risk management, information security and incident response)
6. https://www.nist.gov/cyberframework
   (International framework regarding cybersecurity practices)
7. https://www.nist.gov/cybersecurity-and-privacy
   (general background on cybersecurity practices and risks)

IV. Digital Trust in Technology

Digital trust is about the level of confidence customers have in the digital world, in the services, providers and tools available to them. In the fintech space trust is at the very heart of any relationship as customers are often going to hand over their personal data and their money to a company. Trust can have a direct bearing on customer choice and customer advocacy. With convenience being such a key driver in any purchase, it is unlikely that customers would engage with a business that didn't offer them the level of security and transparency that they required.

To have trust in a fintech company, customers need to believe that they will get the right level of support and that their personal data will not be misused. In many cases, good cybersecurity is a key enabler of the relationship between a provider and a customer as it ensures that the customer's money and other assets are protected. The way a business goes about protecting a customer's data and funds is a key part of the trust equation. However, most customers simply don't have the technical know-how to assess the strength of a fintech company's cybersecurity. In this situation they will be relying on their own intuition, their own research, their brand perception and word of mouth when making a decision about which financial technology provider to go with.

Trust is equally important to the customer when they are choosing between providers, products and services. The privacy aspect of fintech is of course inextricably linked to trust. Customers want to be confident that their personal information will be treated with care and only with their consent. For this reason, many customers will go with the established financial institutions. A company with a good reputation for data protection will have a distinct advantage over a competitor with a poor privacy record. Conversely, customers may be distrustful of a provider that shares their data without their permission even if the provider is offering a better price. Ultimately customer trust in fintech is about accountability and reliability. If something goes wrong and a customer makes a claim, the way that the service is provided during the claims process can be a major factor in how the customer views the company. In other words, the way a company manages a complaint can have a direct impact on the level of trust that the customer has in the organisation. The best approach is for a company to take responsibility for its mistakes and to explain the situation in full to the customer. This is the way in which a company can rebuild customer trust and even turn a negative experience into a positive one.

A BIS (Bank for International Settlements) report provides some interesting statistical data on consumer attitudes to fintech. The 2021 BIS bulletin shows that a significant number of Americans actually expressed greater trust in the established financial institutions, in the fintech companies and in the big tech companies in terms of data privacy. The data is of course only indicative of a general trend as trust levels can vary enormously from one country to another and also from one type of fintech company to another. The report also shows that customers are at least as concerned about price, customer service and transparency as they are about security. It also demonstrates a level of public scepticism about the extent to which big tech companies will share data with third parties. While security is clearly a key factor in the decision to entrust a business with your money, it is not necessarily the only factor. Customers will also vary in their level of technical literacy with some having very little or no understanding of how security actually works.

With the growing uptake of embedded finance in the UAE, trust is likely to be a key differentiator between similar services offered by different providers. So whilst security is a vital component of the trust equation, it is not the only one. In reality, security measures on their own are not sufficient. Customers need to have the confidence that they will get the level of support, reliability, accountability and transparency that they expect. Security and privacy are fundamental to customer trust in fintech but they must be supported by a strong reputation, accountability, compliance with regulation and consumer protection as well as other factors.

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8. https://www.bis.org/publications/bulletin-42-whom-do-consumers-trust-their-data-us-survey-evidence

   (The survey concerns US households, not UAE residents. It just to support a general discussion of trust, and does not prove what UAE consumers think)

V. The UAE's Regulatory and Digital Environment

The United Arab Emirates is a suitable setting for research into embedded finance since digital services play a significant role in its evolving financial environment. People are able to use digital platforms to make payments, access banking, shop and obtain other services. When these services become more interconnected, it becomes necessary to have regulation and technical safeguards in order that innovation may progress hand in hand with adequate protection for users.

The financial environment in the UAE is made up of various regulatory authorities and legal frameworks. The Central Bank of the UAE has responsibility for banking and other financial activities within its area of jurisdiction. The country also has financial free zones which have their own sets of regulations, such as the Abu Dhabi Global Market and the Dubai International Financial Centre. The specific requirements will vary according to the organisation, the nature of its activities, and the jurisdiction in which it is operating. It is therefore incorrect to suppose that a single set of regulations applies in the same way to every financial technology business in the UAE.

A major development is the UAE Open Finance framework, which enables the secure sharing of financial information and, where allowed, the initiation of financial transactions via authorised participants. It makes it possible for customers to access services that make use of data from various financial providers and could thus contribute to the creation of new financial products.

The Open Finance Regulation issued by the Central Bank of the UAE lays down the requirements for organisations that wish to take part and for the way in which the framework operates. It provides for an API Hub and a Trust Framework in order to enable secure interactions between the various participants. The regulation also covers user consent, authentication, secure communication, technology risk management and cybersecurity. These features are concerned with digital trust when different organisations are exchanging financial information, it is necessary for them to have methods of determining which participants are authorised and of ensuring that communication takes place securely. Having shared technical requirements can help to achieve consistency, and suitable authentication together with access controls can lower the chance of unauthorised activity. Yet the success of such arrangements relies on how they are put into effect and kept up.

User consent is also a crucial element of the regulatory approach. The Open Finance Regulation contains requirements relating to explicit consent for the relevant data processing and provides instructions regarding services falling within its scope. This shows that the sharing of financial information should be linked to the relevant purpose and to user authorisation rather than allowing unrestricted access to customers' data.

Financial organisations must identify the risks involved, define the responsibilities, put in place appropriate controls, and be ready for security incidents. they also need to have procedures for dealing with problems and for restoring services. These responsibilities cannot be met by relying on a single security tool. The personal data protection legislation is also part of the UAE's legal environment.

The Federal Decree-Law No. 45 of 2021 lays down a general framework for personal data protection within its provisions. When assessing its application, due account must be taken of the relevant exceptions and of any other rules that might apply to specific financial institutions or free zones. For that reason, research should be based on official legislation and regulatory guidance and not assume that the same requirements apply to all organisations. Even though regulations form a basic basis, the fact that rules exist does not mean that all services are secure or that all customers understand how their information is being handled. Proper resources, staff training, technical protections, and periodic reviews are all necessary for implementation. Providers should also take into account the responsibilities of the third parties who are involved in providing their services.

Another challenge consists in conveying technical and legal information to ordinary users. A privacy notice might include all the necessary details but could still be hard to understand. Since customers will not be able to tell what information is being shared, why it is needed, or which organisation is receiving it, they will have difficulty in making informed decisions. Clear communication must therefore go hand in hand with formal compliance. Technology also undergoes change over time, since new products and business models may bring about risks which were not apparent at the time a service was originally developed. Providers should examine their systems whenever services change, and regulators must keep on taking into account new developments and have to set out their expectations where appropriate. The regulatory environment in the UAE is suitable for looking at the connection between innovation, cybersecurity, and privacy; the Open Finance framework is a specific example of how regulation can deal with issues of data sharing, authentication, and

technology risk management. Whether or not these measures lead to increased public confidence will depend on the way that the providers apply the rules, communicate with users, and respond to their concerns.

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9. https://www.uaelegislation.gov.ae/en/legislations/1972/download *(UAE's federal personal data protection law)*

VI. Case Studies and Practical Applications

Imagine you open up your banking application software on your laptop or smartphone, and you are unable to access your finances. Your purse or wallet equivalent may still show a healthy balance, but the company that is supposed to safeguard your deposits has entered receivership and no one seems to be able to account for where all your money has gone.

Now, jump to reality: financial scandals involving mismanagement of customer deposits are not a figment of our imaginations. They happen, with devastating consequences to the affected parties.

The concept of embedded finance seeks to allow consumers to use banking products like digital wallets, credit, instalments and insurance through third-party applications, rather than through traditional banking channels. Embedded finance products include wallet integrations for online retailers, instalment purchase plans for manufacturers or lenders, and insurance products that are available in business applications.

A financial scandal can erupt when the chain breaks at any point, leaving consumers stranded. The cases below are on how financial scandals, investigations and regulatory issues can implicate embedded finance products.

Case Study 1: United States: Synapse Financial Crisis (2024)

The problem: Many customers were left stranded after the fintech technology company filed for bankruptcy.

Synapse Financial Technologies was a financial technology company that provided intermediary services between fintech companies and banks to facilitate customer banking transactions.

Essentially, when a fintech company partnered with Synapse, its consumers could use a bank-like interface to manage their finances, instead of relying on the services of a traditional bank. When Synapse filed for bankruptcy in 2024, customers of several affiliated fintech applications were left stranded, unable to access their finances. Over 100,000 consumers were affected by the financial scandal. There was also an observed disconnect between the money that appeared on customer-facing applications and the actual cash that should have been on deposit.

To put it another way, consumers could see the money, but not access it, and it was unclear to everyone just how much there was. It is important to emphasise that the mere fact that a disconnect existed did not necessarily imply that the funds had been siphoned off. The issue was not limited to consumer confusion banks faced difficulties in determining how much money should be credited to the customers' accounts, owing to the disconnect between the customer-facing and bank-facing books.

Why does this matter for embedded finance?

Synapse is a cautionary tale about the dangers of embedded finance. Essentially, a single financial technology company was responsible for facilitating financial transactions between banks and consumers.

While the company itself was blameless in the eyes of the law, the burden of proof of misappropriation of funds was on the plaintiffs, the fallout was severe for the consumers, who were left stranded.

Lesson: Financial technology companies must keep their customer-facing application books in order and ensure that they correspond with the actual cash on hand at any given time. They should establish accountability as to which entity has custody of customer deposits.

While it might be inconvenient, consumers should not be caught up in a financial and legal nightmare if they wish to retrieve their hard-earned money.

Source:

https://www.consumerfinance.gov/enforcement/actions/synapse-financial-technologies-inc/

https://dfpi.ca.gov/press_release/dfpi-revokes-finance-lending-license-of-synapse-credit-llc/

Case Study 2: Germany: Wire card Scandal (2020)

The problem: €1.9bn that appeared to exist, did not.

Wire card was a payment processing company that specialised in processing electronic transactions for businesses. As a publicly-traded company, Wire card made promising statements about its stature in the payments space and its technological capabilities.

In June 2020, however, Wire card announced that approximately €1.9bn that was supposed to exist in trustee accounts did not, in fact, exist. As a result, the company was declared insolvent and investigations ensued, with allegations of fraud being levelled at the company.

So why does this matter for embedded finance?

Again, Wire card is not an embedded finance company, so it would be incorrect to categorise it as such. It does serve as a salient reminder that a company's stature in the financial space and its image do not always reflect its financial realities on the ground. Consumers or businesses should not trust a company simply because it claims to be trustworthy, due diligence on financial matters, on the part of consumers and businesses, is critical.

Lesson: Wirecard shows us that businesses need to ensure that their books are in order and that there is no scope for fraud. Independent audits and regulatory oversight are also necessary to prevent, detect and deal with financial mismanagement.

Source:

https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/659639/IPOL_BRI%282021%29659639_EN.pdf

https://www.europarl.europa.eu/news/nl/press-room/20210319IPR00444/wirecard-meps-call-for-new-audit-rules-and-whistle-blowers-protection-a

Case Study 3: International A7 Payment Network Allegations (Reported in 2026)

The problem: How can seemingly mundane-looking payments be used to facilitate money laundering?

According to a report in 2026, there is an alleged international network of companies tied to A7 that allegedly laundered more than $6.9bn through the banking system, using front companies, fake invoices, and forged documents to conceal transactions and evade sanctions.

This report shows us that allegedly, businesses both in the UAE, Hong Kong and the UK have taken advantage of the complexities of international payments to facilitate financial crime and evade sanctions.

This case is not directly applicable to embedded finance. the allegations relate to payments processing and not to a product that facilitates banking products within a business application. At the same time, the report highlights the potential that financial technology and the embedded finance space have for misuse by bad actors.

Lesson: Businesses need to have appropriate customer verification and transaction monitoring procedures in place to combat financial crime. These procedures need to be applied globally, since transactions could originate from one jurisdiction and terminate in another, passing through several financial intermediaries along the way.

Source:

https://www.ft.com/content/1273931a-238d-45a9-85b3-5c847e79b5ea

Case Study 4: Netherlands: Modulr Finance B.V. (2026)

The problem: A company failed to implement appropriate checks and monitoring procedures.

On 5 October 2026, De Nederlandsche Bank issued an administrative fine of €722,160 to Modulr Finance B.V. for deficiencies that were discovered in the area of customer due diligence and ongoing anti-money laundering monitoring in the period from January 2023 to July 2024.

Customer due diligence refers to the procedures that a business must implement to establish appropriate risk management measures with regard to its customers. Ongoing monitoring of customers ensures that businesses are aware of any changes in the customer's risk profile.

These steps are necessary because businesses can be used to facilitate money laundering and other financial crimes by persons with malicious intent.

This case differs from the Wirecard and Synapse scandals, in that it is not an allegation of misappropriation of customer funds. The administrative penalty was issued because Modulr Finance BV failed to implement appropriate procedures for customer due diligence and ongoing monitoring.

Why does this matter for embedded finance?

This case is important because it underscores the potential for failure in customer due diligence and transaction monitoring procedures in the context of embedded finance. Modular and similar financial technology start-ups operate in the grey space between banks and businesses. With the former providing financial services and the latter providing technological infrastructure, responsibility for customer due diligence and transaction monitoring is not clear-cut.

Lesson: Appropriate identity checks and monitoring procedures are vital. They should be implemented not as a one-time procedure, but as an ongoing process. This ensures that businesses are not caught off-guard in the event of a financial crime.

Source:

https://www.dnb.nl/en/general-news/enforcement-measures-2026/administrative-fine-imposed-on-modulr-finance-b-v-for-inadequate-customer-due-diligence/

What Do These Cases Say About Embedded Finance?

These cases, although different, underscore several systemic issues that are relevant to the world of embedded finance. As embedded finance products allow consumers to use banking products through non-traditional channels, businesses involved in embedded finance should take note of the following lessons:

1. **Customer funds must be accounted for.** The Synapse case is a reminder that the money that appears on the consumer-facing application must reflect the actual money that is on deposit. Regular book reconciliations are therefore necessary to prevent disconnects between the consumer-facing and bank-facing books.

2. **A good reputation is not a guarantee of goodness.** Wirecard shows that companies must ensure that their books are in order and that appropriate controls are in place to prevent fraud. Consumers and businesses should ensure that they do not trust a company simply because it claims to be trustworthy.

3. **Speed is not a virtue, unless it is needed.** The A7 case shows that fast payments can be abused to facilitate financial crime. Businesses should therefore implement appropriate transaction monitoring systems to detect and prevent financial crime.

4. **Good checks prevent bad financial crime.** The De Nederlandsche Bank case highlights the need for appropriate checks and ongoing monitoring of transactions to detect and prevent financial crime.

VII. Discussions and Recommendations

The research shows that both cyber security and data privacy are foundations of digital trust in embedded finance. Connected financial services can give benefits in convenience but rely on many systems and organisations. A lack of access controls, unclear data sharing rules or a weak incident response can all hurt consumer confidence. Many security controls can lower risk, such as multi-factor authentication, secure API design, encryption, access controls, regular software updates and an incident response plan. No single control can remove all risk.

Cyber security is a duty and providers must check cyber security posture as services and risks change. Data privacy is equally important to cyber security because data privacy is not about stopping unauthorised access. Providers must explain why data is collected limit data processing to the purposes for which data is collected and state any data sharing that happens. The UAE Open Finance framework shows rules that cover consent, authentication, secure communication and technology risk management and from this several recommendations arise.

First, financial service providers should use security controls that match the cyber security risks of the services they give. Providers should keep checking APIs, access controls, software updates and third-party arrangements. **Second**, privacy notices and consent procedures should be written in language.

Customers must understand what data is shared why data is shared and with whom data is shared. **Third**, organisations must have complaint and incident reporting steps. If something goes wrong customers must know who to contact and what steps can fix the problem. **Fourth**, providers and regulators should raise consumer awareness. Consumers would gain from spotting phishing messages protecting their accounts and checking the legitimacy of requests for data.

The responsibility of security should not fall on the customer. These recommendations come with the limits of the research. This study did not test platforms or measure public views in the UAE. More research could use a well-designed survey of UAE consumers or interviews, with professionals with proper consent and academic rules. In short embedded finance can grow responsibly when innovation rests on systems right data practices, accountability and transparency. These parts can help build trust but will not guarantee it.

VIII. Conclusion

So, this study analysed how embedding cybersecurity and data privacy in embedded finance can help raise digital trust for the future in the UEU. It examined how embedded finance works, the risks associated to interlinked financial systems, the need for privacy of personal data and the importance of having a suitable regulatory framework in the UAE.

The analysis reveals that cybersecurity and data privacy are highly relevant because embedded finance may require a few organisations and systems to operate in concert. While proper authentication, secure communications, access controls and careful design of APIs will not remove or eliminate security risks completely along with an effective incident response. Massively, defined consent processes, legitimate data accumulation, and transparency related to protection heaps can be for users in seeing how their information will be made use of.

The UAE Open Finance framework is a live example of how the regulation can address concepts such as secure data sharing, data authentication, consent and technology risk management. The existence of regulatory requirements does not mean that all services are secure or that all users understand such practices.

Hence, the research concludes that behaviour towards digital trust requires more than terms of technical protection related to accountability but require

accountably, transparency, trustful services and an effective consumer protection regulatory regime. However, if financial technology providers pair the right protection and controls with meaningful service design, it is possible to innovate with a genuine commitment to users' security and privacy. Ultimately, the fate of embedded finance in the UAE rests, to some extent, on whether its potential betterment and convenience are matched by an equally responsible approach to protecting customer information.

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